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System Overview

The system is an integrated hybrid e-commerce platform that enables new brands to launch online stores quickly while also helping them attract customers. It contains website creation, order management, and traffic support in a single solution, removing the need to rely on multiple disconnected services. This reduces costs, prevents data inconsistencies, and speeds up the time required for brands to enter the market.

The main users of the system are **merchants**, who manage their stores and products, and **shoppers** who browse and purchase items. Other stakeholders include logistics providers that handle delivery.

The motivation behind the system is to lower barriers for emerging businesses to start selling online efficiently. It is designed to support different e-commerce models from general marketplace like Amazon and Jumia to specialized stores.

Product Backlog

Customer Stories

1. As a customer, I want to create an account so that I can track my orders.

Acceptance Criteria:

- Given I, the customer, am on the registration page, when I enter a valid email and password, then my account is created successfully.
- Given I, the customer, am on the registration page, when I enter invalid credentials, then I see a validation error message.
- Given I, the customer, complete registration, when the process finishes, then I see a confirmation message.

2. As a customer, I want to browse products by category, so that I can easily find what I need.

Acceptance Criteria:

- Given I, the customer, am viewing categories, when I select a category, then I see only products in that category.
- Given I, the customer, am on a category page, when products load, then they are grouped correctly.
- Given I, the customer, select a category, when the page loads, then it loads within 3 seconds.

3. As a customer, I want to search for products so that I can quickly locate specific items.

Acceptance Criteria:

- Given I, the customer, am using the search bar, when I enter a keyword, then I see matching results.
- Given I, the customer, search for a product, when no match exists, then I see a “No Results Found” message.
- Given I, the customer, perform a search, when results are displayed, then they appear within 2 seconds.

4. As a customer, I want to add products to a cart so that I can purchase multiple items at once.

Acceptance Criteria:

- Given I, the customer, am viewing a product, when I click “Add to Cart”, then the cart count increases.
- Given I, the customer, open my cart, when items exist, then I see correct product details.
- Given I, the customer, update quantity, when I change it, then the total price updates automatically.

5. As a customer, I want to check out securely so that my payment information is protected.

Acceptance Criteria:

- Given I, the customer, am on the checkout page, when I enter shipping and payment details, then they are accepted.
- Given I, the customer, complete payment, when processed, then it is handled securely via HTTPS.
- Given I, the customer, finish checkout, when payment succeeds, then I see an order confirmation page.

6. As a customer, I want to view product reviews so that I can make informed decisions.

Acceptance Criteria:

- Given I, the customer, am on a product page, when reviews exist, then I see the average rating.

- Given I, the customer, open reviews, when they are available, then I can read all reviews.
- Given I, the customer, attempt to submit a review, when I am not a verified purchaser, then I am prevented..

7. As a customer, I want to track my order status so that I know when it will arrive.

Acceptance Criteria:

- Given I, the customer, am on my profile, when I open order history, then I see my past orders.
- Given I, the customer, view an order, when it loads, then I see its status.
- Given I, the customer, have an order in progress, when its status changes, then it updates in real time.

Seller Stories

8. As a seller, I want to create a seller account so that I can list my products.

Acceptance Criteria:

- Given I, the seller, am on the registration page, when I submit business information, then my application is sent for review.
- Given I, the seller, have applied, when the admin reviews it, then I receive approval or rejection notification.
- Given I, the seller, am approved, when the process completes, then my account becomes active.

9. As a seller, I want to upload product details so that customers can purchase them.

Acceptance Criteria:

- Given I, the seller, am adding a product, when required fields are missing, then submission is rejected.
- Given I, the seller, enter valid product details, when submitted, then it appears in my dashboard.
- Given I, the seller, submit a product, when approved, then it becomes visible to customers.

10. As a seller, I want to update stock levels so that customers do not order unavailable items.

Acceptance Criteria:

- Given I, the seller, am managing products, when I update stock quantity, then the new value is saved.
- Given I, the seller, set stock to zero, when the update is applied, then the product shows "Out of Stock."
- Given I, the seller, update stock, when saved, then it reflects immediately on the product page.

11. As a seller, I want to see sales analytics so that I can monitor performance.

Acceptance Criteria:

- Given I, the seller, am on my dashboard, when I open analytics, then I see total revenue.
- Given I, the seller, view analytics, when data loads, then I see number of items sold.
- Given I, the seller, check analytics, when updated, then data reflects daily changes.

12. As a seller, I want to request a custom store extension so that I can grow my brand independently.

Acceptance Criteria:

- Given I, the seller, want an upgrade, when I submit a request form, then it is sent to admin.
- Given I, the seller, submit a request, when processed, then admin receives notification.
- Given I, the seller, check status, when viewing dashboard, then I see upgrade status..

13. As a seller, I want my store extension to have my own branding, so that my business looks professional.

Acceptance Criteria:

- Given I, the seller, am customizing my store, when I upload a logo, then it is applied.
- Given I, the seller, choose a theme, when selected, then it becomes my store's theme.
- Given I, the seller, apply branding changes, when saved, then they reflect on my store page.

14. As a seller, I want to manage orders so that I can process shipments efficiently.

Acceptance Criteria:

- Given I, the seller, am on my dashboard, when I view orders, then I see incoming orders.
- Given I, the seller, update an order status, when I change it, then it is saved.
- Given I, the seller, update order status, when completed, then customers receive notification.

15. As an admin, I want to approve sellers so that only verified businesses sell on the platform.

Acceptance Criteria:

- Given I, the admin, am reviewing applications, when I view pending sellers, then I see all applications.
- Given I, the admin, review an application, when I approve or reject, then the decision is applied.
- Given I, the admin, make a decision, when completed, then it is logged in the system.

16. As an admin, I want to manage commission rates so that the platform generates revenue.

Acceptance Criteria:

- Given I, the admin, am configuring settings, when I set commission percentage, then it is saved.
- Given I, the admin, view transactions, when a sale occurs, then commission is calculated automatically.
- Given I, the admin, update commission, when applied, then it affects future transactions only.

17. As an admin, I want to monitor transactions so that fraud can be detected.

Acceptance Criteria:

- Given I, the admin, am on the dashboard, when I view transactions, then I see the full history.
- Given I, the admin, monitor activity, when suspicious behavior occurs, then it is flagged.
- Given I, the admin, filter transactions, when I apply filters, then results are refined.

18. As an admin, I want to manage seller upgrade subscriptions so that sellers can pay for their independent website extension.

Acceptance Criteria:

- Given I, the admin, manage subscriptions, when I activate or deactivate one, then the status updates.
- Given I, the admin, view subscription data, when I check payment status, then it is recorded.
- Given I, the admin, manage subscriptions, when one expires, then the store extension is disabled.

System Non-Functional-Related Stories

19. As a user, I want the website to load quickly, so that I have a smooth experience.

Acceptance Criteria:

- Given I, the user, access the website, when pages load under normal traffic, then they load within 3 seconds.
- Given I, the user, use the system, when 500 users are active, then the system remains stable.

- Given I, the user, perform checkout, when processing occurs, then no timeout errors happen.

20. As a user, I want my data to be encrypted so that my personal information is secure.

Acceptance Criteria:

- Given I, the user, use the system, when data is transmitted, then it is encrypted via HTTPS.
- Given I, the user, create an account, when my password is stored, then it is encrypted.
- Given I, the user, attempts unauthorized access, when detected, then it is logged by the system.

Functional And Non-Functional-Requirements

Story 4: Add to Cart

Functional Requirements:

- The system shall allow customers to add products to a shopping cart.
- The system shall calculate the total cart value automatically.
- The system shall allow quantity modification.

Non-Functional Requirements:

- Cart updates shall reflect within 1 second.
- Cart data shall persist if the user refreshes the page.

Story 5: Secure Checkout

Functional Requirements:

- The system shall process payments via a secure gateway.
- The system shall generate an order confirmation.

- The system shall store order history.

Non-Functional Requirements:

- Payment data must be encrypted (SSL/TLS).
- The checkout process must not exceed 5 seconds of processing time.

Story 12: Store Extension Request

Functional Requirements:

- The system shall allow sellers to submit upgrade requests.
- The system shall process subscription payments.
- The system shall generate a unique subdomain.

Non-Functional Requirements:

- Subdomain activation within 24 hours.
- System uptime must be 99 percent.

Story 15: Admin Approves Sellers

Functional Requirements:

- The system shall allow the admin to approve or reject sellers.
- The system shall notify the seller of the decision.
- The system shall log approval actions.

Non-Functional Requirements:

- Only authorized admin users can access the approval panel.
- The approval system must maintain audit logs.

Story 19: Website Performance

Functional Requirements:

- The system shall monitor server performance.
- The system shall alert the admin to downtime.

Non-Functional Requirements:

- The website must load within 3 seconds.
- System availability must be 99 percent monthly uptime.

Requirements Traceability Matrix

Requirement ID	Category	Requirement Description	Status	Design Element (TBD)	Test Case ID (TBD)
REQ-C01	B2C Consumer	The system shall allow users to register and authenticate via email or social login.	Identified	Django Auth Framework	
REQ-C02	B2C Consumer	The system shall provide a global search bar to query products across all local brands.	Identified		
REQ-C03	B2C Consumer	The system shall allow users to filter products by category, price, and brand.	Identified		

REQ-C04	B2C Consumer	The system shall feature a unified shopping cart allowing users to purchase items from multiple local brands in a single transaction.	Identified		
REQ-C05	B2C Consumer	The system shall generate a unique tracking number and order history for every consumer purchase.	Identified		
REQ-B01	B2B Vendor	The system shall allow local brands to register for a vendor account.	Identified		
REQ-B02	B2B Vendor	The system shall implement a subscription billing module to process automated monthly payments for vendors.	Identified		
REQ-B03	B2B Vendor	The system shall provide vendors with an isolated dashboard to manage their specific inventory, pricing, and orders.	Identified		
REQ-B04	B2B Vendor	The system shall allow vendors to customize their storefront's basic UI elements (e.g., logo, brand colors, banner).	Identified		
REQ-S01	System Core	The system shall cache frequently accessed data (like homepage products) to maintain page load speeds under 2 seconds.	Identified		

REQ-F01	Frontend / UI	The system's user interface shall be fully responsive, rendering correctly on mobile, tablet, and desktop viewports.	Identified		
REQ-F02	Frontend / UI	The B2C marketplace frontend shall update the shopping cart item count and total price dynamically without requiring a full page reload.	Identified		
REQ-F03	Frontend / UI	The marketplace homepage shall utilize infinite scrolling or dynamic pagination for browsing the global product catalog.	Identified		
REQ-F04	Frontend / UI	The frontend shall display skeleton loaders or visual progress indicators for any data retrieval taking longer than 500ms.	Identified		
REQ-F05	Frontend / UI	The B2B vendor dashboard shall provide a real-time visual preview of storefront changes as the vendor customizes their layout and colors.	Identified		
REQ-F06	Frontend / UI	The vendor storefront UI shall support drag-and-drop functionality for uploading product images.	Identified		

REQ-F07	Frontend / UI	The platform's frontend shall comply with basic web accessibility standards (WCAG), including proper contrast ratios and alt-text for product images.	Identified		
REQ-F08	Frontend / UI	Form inputs (e.g., checkout, registration) shall provide real-time inline validation errors before the user submits the form.	Identified		

Risks

- Payment and shipping: the payment demo was done just to show a layout of how a payment might work and is not actually connected to a payment service like Visa or Mastercard, only showing a successful or denied payment.
- Data consistency : we have to ensure the data syncing for the standalone stores that purchase the subscription of being independent in the market place (Inventory, orders, status)
- Time risk: building a customized market place in this short period can cause incomplete features and less quality due to the short time at hand
- Operational risk: the point of an agreement with an outside source for all deliveries might cause problems for sellers who value simplicity rather than achieving the highest profits and discourage them due to minute knowledge about operational services.
- Privacy risks: the simplistic log in system might cause problems if a user tries logging in as another user seeing data that isn't theirs to see and can add inputs that might harm the actual seller for these products giving a weak sense of security.

Features to Be Completed

We will focus on building the necessary features required to cover the core transaction lifecycle. Below, we list each of the features according to their stakeholders as well as the reason for prioritizing them.

Customer

1. Create Account
 - Needed for authentication and ownership of orders.
 - Without accounts, modeling real users is impossible.

2. Browse Products by category
3. Search Products
 - Essential to support the user navigation experience
4. Shopping Cart Functionality: Adding, removing, and updating¹ products
 - Required to model session state and transaction preparation
5. Secure Checkout
 - Required to complete the purchase lifecycle.

Seller

1. Create Seller Account
 - Needed to distinguish between sellers and customers
2. Upload Product details
 - Basic requirement for sellers to offer their products
3. Update Stock Levels
 - Required to prevent unrealistic ordering
 - Maintains consistency between orders and inventory
4. Manage Orders
 - Sellers must see orders to process them
 - Needed to complete the seller side of the purchase lifecycle

¹ By “updating”, we mean updating the count of each item

WBS Diagram

<https://drive.google.com/file/d/1fIGDVqXEyx03F-ibEhsOA3B-AKP7gc-e/view?usp=sharing>

User manual

https://drive.google.com/file/d/1oxwxU9A-Nvc_IX0cjsxR7BE2qVKf_NWi/view?usp=drivesdk

Use Cases

https://lucid.app/lucidchart/ef103a6f-6116-4471-bb0b-1fdae94c99d9/edit?invitationId=inv_84391e61-999b-4f3d-a5a5-a4f189d8457c&page=CLp1aG0dCMu6f#

Admin

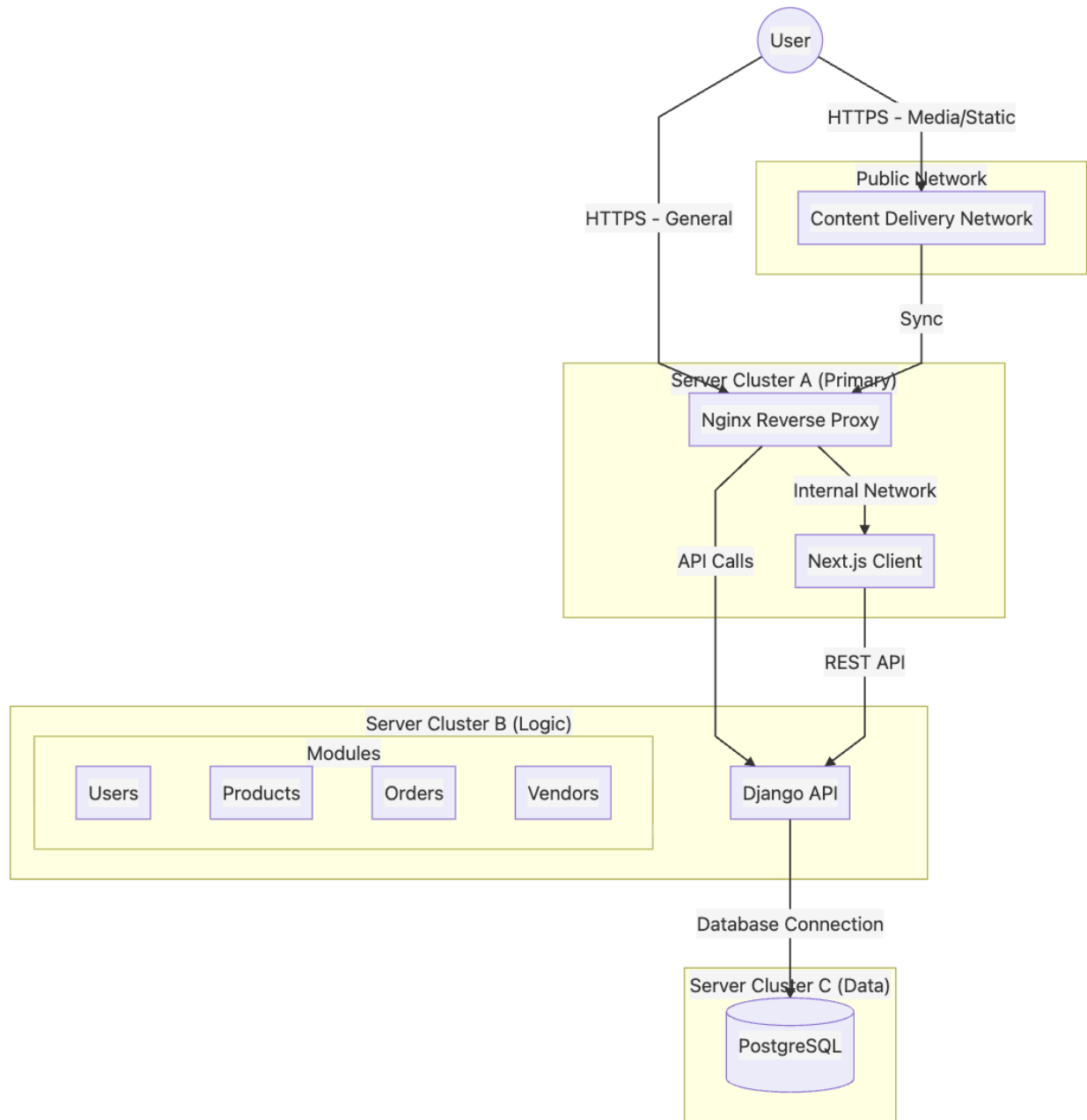
1. Approve Sellers
 - Represents basic moderation workflow
2. Manage Commission Rates
 - Core platform revenue logic

Team Definition of Done (DoD)

A user story / backlog item is Done only when all are true:

- Requirements satisfied
- Code complete
- Quality gates: No critical bugs; no failing checks
- Testing:
- Security & data:
- Deployable: Runs in the staging/demo environment with no manual hacks; demo script (if needed) prepared.

Architectural Design



2. Architectural Patterns

Client-Server Architecture

The system follows a classic **Client-Server** pattern:

- **Client:** A modern Next.js frontend providing a responsive and dynamic user experience.
- **Server:** A robust Django backend exposing RESTful APIs for business logic and data management.

Layered Architecture (Backend)

The backend is organized into a **Layered Monolith**:

1. **Presentation Layer:** REST API endpoints for communication with the frontend.
2. **Business Layer:** Modularized logic for **users**, **products**, **orders**, **vendors**, and **subscriptions**.
3. **Database Layer:** PostgreSQL for reliable and ACID-compliant data storage.

3. Component Interaction

- **Nginx:** Acts as a reverse proxy, routing traffic to the frontend and backend services.
- **Docker Compose:** Orchestrates the deployment of the frontend, backend, and database containers, ensuring environment consistency.

4. Non-functional Requirements

Security

The system implements **Layered Security**:

- **Authentication:** JWT-based or Session-based authentication for sensitive endpoints.
- **Nginx Proxy:** Hides backend implementation details and provides a single entry point.
- **Django Security:** Built-in protections against CSRF, XSS, and SQL Injection.

Performance

- **CDN (Content Delivery Network):** Static assets (JS, CSS, Images, Product Media) are cached and served via a global CDN to minimize latency for users across different regions.

- **Nginx Caching:** Acts as an additional cache layer for frequently accessed API responses.

Maintainability

- **Fine-grained Components:** The backend is divided into self-contained Django apps (e.g., **orders**, **vendors**), making it easy to isolate and update specific features.
- **Containerization:** Simplifies dependency management and deployment across different environments.

5. Server Distribution & Scalability

To ensure **Availability** and **Scalability**, the architecture supports distribution across multiple servers:

- **Distributed Deployment:** Each major component (Frontend, Backend, Database) can reside on different physical or virtual servers to prevent a single point of failure.
- **Microservice Ready:** As core backend modules (like **Orders** or **Vendors**) grow, they can be extracted into independent services running on dedicated infrastructure, as outlined in the Microservices pattern.
- **Redundancy:** Critical components like the Database can be replicated across servers to ensure the "uptime" required for enterprise-grade applications.

6. Future Roadmap

As the platform scales to accommodate a larger number of users, we plan to transition towards a **Microservices Architecture**. This will involve decoupling core services (like **payments** or **inventory**) into independent, fault-tolerant units, as recommended in the architectural design principles.

